Deepfake Detection & Authenticity Verification using Artificial Agentic Generative Quantum Intelligence , Cryptographic authentication and AI fingerprinting

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Abstract - Deepfake detection involves verifying the authenticity of visual and audio media using AI technologies, including machine learning, forensic analysis, and biometric comparison. The emergence of deepfakes challenges digital trust and necessitates innovative detection approaches. This paper critiques traditional methods, proposing a new framework for Deepfake Detection and Authenticity Verification utilizing Artificial Agentic Generative Quantum Intelligence (AAGQI). This framework enhances authenticity verification through real-time analysis of various signals and employs quantum-inspired intelligence for improved pattern recognition. It also incorporates cryptographic authentication and AI fingerprinting to preserve media integrity and track generative models. The framework shows potential for applications in journalism, legal forensics, and secure communications.

Keywords - Deepfake detection, authenticity verification, agentic AI, quantum intelligence, cryptographic authentication, AI fingerprinting, digital provenance, multimodal forensics.

I. INTRODUCTION

Deepfake detection involves finding and verifying the authenticity of videos, photos, and audio with AI technologies such as machine learning models, forensic analysis, and biometric comparison. The quick rise of deepfakes is a significant threat to digital trust. This requires new methods for detection and verification. Unlike traditional systems that follow static rules, these agentic frameworks utilize multiple expert AI agents to perform multi-dimensional forensic investigations in real-time. Deepfakes, created using generative adversarial networks (GANs) and diffusion models, produce hyper-realistic media. They present security risks in finance, governance, and media. This shows the weaknesses of traditional AI detectors against new adversarial methods. This highlights the need for improved systems that use quantum technology with autonomous adaptability. Previous studies focused on spotting anomalies, but now the approach emphasizes training AI to identify AI-generated content. To tackle deepfakes effectively, the strategy should change to verifying their origins. This requires establishing digital provenance. This process includes tracing a file's history, detailing its original creator, the time and place of creation, and any AI modifications, using cryptographic metadata and digital signatures. Deepfake detection utilizing pixel analysis involves identifying subtle digital traces and inconsistencies that may not be noticeable to the human eye. Key techniques for detecting digital manipulation include Error Level Analysis (ELA) for checking pixel degradation. Pixel-wise Temporal Frequency analysis looks for flickering in video using Fourier transforms. Color Space Analysis uses different spaces like YCbCr to show variations. Steganalysis-based Models focus on pixel-level changes to spot high-frequency patterns from AI up-sampling. The existing architecture is shown in Fig. 1 **[1]** which defines a new forgery detection method using pixel-wise temporal frequency is introduced. It shows how temporal frequency successfully identifies forgery, especially in situations where traditional methods may not work. Unlike spatial frequency, this method can reveal local temporal inconsistencies, making it easier to detect general deepfake videos.

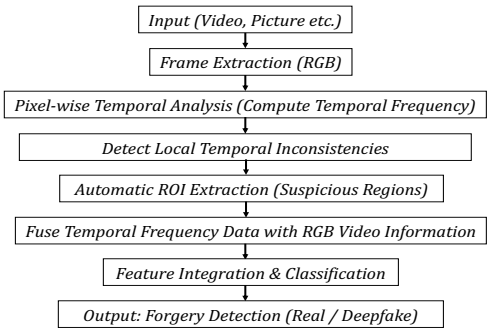

Fig. 1: Architecture of the existing system

This paper introduces a framework for Deepfake Detection and Authenticity Verification using Artificial Agentic Generative Quantum Intelligence (AAGQI) which emphasizes proactive authenticity management through AI analysis of multimedia signals, quantum recognition, cryptographic media provenance authentication, and AI fingerprinting, thereby enhancing detection and attribution capabilities while addressing risks in the sensitive The work introduces a multi-layered framework combining the latest technologies like Agentic AI, Quantum Computing, and Cryptographic standards.

II. REVIEW OF LITERATURE

The rapid growth of generative AI, which is transforming the creation of text, images, and videos, has led to concerns regarding authenticity, misinformation, and academic honesty. As generative systems like ChatGPT gain popularity, there is a pressing need for researchers and educators to develop effective methods for detecting AI-generated content.

A. Detection of AI-Generated Text

AI-generated text has smooth language, correct grammar, and consistency, but it often misses depth and personal stories. This can create a disconnect from real-world context. Common qualities include awkward comparisons and repeated structures. A significant problem is "hallucinations," where AI may generate false information. This emphasizes the need to check content against reliable sources.

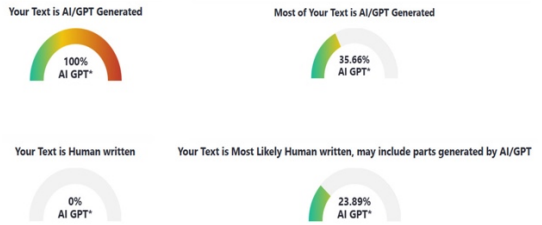

Fig. 2: AI-Generated Text

B. Detection of AI-Generated Images and Videos

Scholars are broadening their examination of synthetic media to include visual and audiovisual elements alongside text. Key indicators of AI-generated images include anatomical irregularities, background inconsistencies, and lighting flaws. For AI-generated videos, signs involve unnatural movements, infrequent blinking, stiff gestures, and lip sync errors, complemented by audio features such as monotone voices and speech delays.

Fig. 3: AI-Generated Image

C. Tools and Verification Approaches

i) Past Deepfake Detection Methods (2017–2020)

a) General Approach

Günay et al. **[2]** proposes a multi-dimensional framework for evaluating Deepfake attacks as cyber-enabled crimes, addressing the gap in unified structures previously identified in literature. It supports strategic, regulatory, and law enforcement efforts while enhancing academic discourse. The ten-dimensional model includes factors such as content type, intent, target, attacker, timing, duration, distribution channel, quality, interaction type, and type of harm.

b) Eye-Blink Detection (2018) **[3-6]**

In this study **[3]**, a novel method for deepfake detection based on human eye blink patterns is proposed, emphasizing the challenges of imitating these patterns in synthetic videos which analyzes individual blink characteristics influenced by genetics and muscle tone, employing computer vision and machine learning for pattern extraction for identifying deepfakes. *T. Jung et al.* **[4]** proposed DeepVision which is a new algorithm for detecting Deepfakes generated by GANs, focusing on changes in blinking patterns affected by physical, cognitive, and emotional factors. Using a heuristic approach that combines insights from medicine, biology, and machine learning, it verifies video integrity by monitoring eye blink timing and frequency achieving 87.5% accuracy, thus overcoming traditional pixel-based verification limitations. *Li, Yuezun et al.* **[5]** presented a novel method for detecting fake face videos produced by neural networks, centering on the detection of eye blinking, which is often absent in synthetic videos. Our approach performs well on eye-blinking detection benchmarks and effectively identifies videos generated by DeepFake technology. *H.K Verma et al.* **[6]** introduced a hybrid method for detecting deepfake videos that integrates human eye blinking pattern analysis, Xception image features, and a Bidirectional Long Short Term Memory network (Bi-LSTM) which employs Convolutional Neural Networks for image feature extraction and Bi-LSTM for sequential analysis, utilizing a spatio-temporal attention mechanism to focus on significant frames.

c) Head Pose & Facial Landmark **[7-8]** and Frequency Domain Analysis

Lutz et al. **[7]** investigates a DeepFake detection method reliant on head pose estimation, applicable to images generated via autoencoder-based face swapping. While literature praises this approach as effective, a reproducibility study reveals its claimed advantages are overstated. *Federico Becattini et al.* **[8]** looks into how Head Pose Estimation (HPE) patterns can help detect deepfakes, using algorithms like FSA-Net, SynergyNet, and WSM. When

evaluated with K-Nearest Neighbor and Dynamic Time Warping, these methods show that FSA-Net excels at feature extraction, reaching 86% accuracy in identity swap tasks and 86.5% in expression swap tasks on the FaceForensics++ dataset.

ii) Present Deepfake Detection Methods (2021–25)

a) Deep Learning-Based Detection **[9-12]**

Bisme AS et al. **[9]** presents a hybrid deepfake detection model that combines Convolutional Neural Networks (CNN) and Multi-Layer Perceptron (MLP) to improve detection accuracy. The model uses facial landmarks for feature extraction to identify geometric inconsistencies and facial distortions which shows that the hybrid model reaches 98.99% accuracy, precision, and F1-score, surpassing traditional CNN and Long Short-Term Memory (LSTM) networks. *Carter B et al.* **[10]** introduces a new way to detect deepfakes using only facial landmarks which focuses on spotting small inconsistencies in facial movements instead of just processing raw images. Different neural network models showed significant accuracy: RNN models had 96% accuracy, ANN models achieved 93%, and CNN models came in at around 78%.

T. S, T. Babu et al. **[11]** reviews deepfake detection technologies, emphasizing traditional machine learning, deep learning, and preprocessing techniques. It evaluates various neural network models, including MesoNet, Xception, CapsuleNet, and classical CNNs like DenseNet and ResNet. Additionally, it examines preprocessing methods, such as YCbCr color transformation and edge detection, to enhance accuracy despite advancements in generative adversarial networks (GANs). *Dave et al.* **[12]** proposed a new deepfake detection system that uses SVMs and CNNs which achieves accuracy rates of up to 88.33% in detecting deepfakes in images.

b) Transformer-Based Models

To address both spatial and temporal inconsistencies, *Wu J et al.* **[13]** suggested a two-stage network, this network includes a Mask-Guided Local Attention module (MLA) to identify unique local patterns in PPG maps, Temporal Transformer to connect features from nearby PPG maps over long distances. When the extensive testing was performed on the FaceForensics + + and Celeb-DF datasets shows that the method outperforms all other rPPG-based approaches.

c) Other Methods **[14-21]**

Y. Ni et al. **[14]** proposed a novel method for extracting remote photoplethysmography (rPPG) signals using Fast Fourier Transform (FFT) in which time-frequency signals are organized into matrices to create Matrix Visualization Heatmaps (MVHM), which train an image classification network through various time-frequency representations and attention mechanisms, we attained a detection accuracy of 99.22% for fake video identification, surpassing existing algorithms and demonstrating the efficacy of FFT and attention mechanisms in fake face detection. Liu, W et al. **[15]** presented a new method for identifying lip forgery by spotting differences between lip movements and the related audio signals. By mimicking human thinking, this approach improves accuracy by considering the biological links between the lips and head.

Tan, C et al. **[16]** presented FreqNet, a new universal deepfake detector designed to identify unseen deepfake images with limited training data. *Hinke-Navarro A et al.* **[17]** introduces a new deep-learning detection framework that combines handcrafted frequency-domain features with standard RGB inputs. This hybrid model uses both frequency and spatial domain artifacts from image manipulation to improve the classifier's performance. The main frequency handcrafted features evaluated include the Steganalysis Rich Model, Discrete Cosine Transform, Error Level Analysis, Singular Value Decomposition, and Discrete Fourier Transform.

Bachir et al. **[18]** evaluates the effectiveness of handcrafted feature-based methods. Six leading descriptors were tested to extract image features, and SVMs were trained on these features to create a detection model. Results reveal that the SIFT descriptor excels with a detection accuracy of 83.50% on videos made with the neural texture technique, outperforming deep learning methods. *M. S. Rana and A. H. Sung* **[19]** demonstrates that traditional machine learning (ML) methods can outperform deep learning (DL) techniques in detecting Deepfakes, achieving accuracy rates of 99.84% on the FaceForensics++ dataset, 99.38% on DFDC, 99.66% on VDFD, and 99.43% on Celeb-DF. According to *M. S. Rana et al.* **[20]**, advances in AI and ML have created multimedia manipulation tools for legitimate purposes like entertainment and education. However, they have also resulted in the misuse of high-quality fake content known as Deepfake for spreading misinformation and harassment. The article **[21]** discusses the benefits, features, applications, and limitations of these detection methods, addressing challenges and predicting future trends. It also offers

recommendations for improving AI's role in deepfake detection to foster a safer digital environment.

D. Surveys and Analysis Reports **[22-26]**

As per the surveys conducted in Pakistan by *M. Usama Tanveer Gujjar et al.* **[22]** and *Qureshi SM et al.* **[23]** categorizes deepfake detection techniques using deep learning, emphasizing their roles in digital watermarking and forensics which reviews methods like CNNs, RNNs, attention mechanisms, and facial geometry models, finding that hybrid architectures such as ResNet-Swish-BiLSTM and spatiotemporal attention networks are particularly effective for detecting altered media. This research systematically surveys digital forensic methods for deepfake detection, evaluating the effectiveness and limitations of various techniques across image, video, text, and audio modalities. It includes case studies of notable incidents to demonstrate real-world performance and discusses research limitations in cross-modality detection. The study concludes with a call for ongoing innovation to address the rapid evolution of deepfake technology, providing a thorough overview for researchers, practitioners, and policymakers.

Other countries also claimed about the same such as *Luan, TAO* **[24]** reviews progress in deepfake detection technology, focusing on traditional and modern methods, especially deep learning techniques. Outlines the principles and classifications of deepfake technology and looks at approaches like physical feature detection, large model methods, and biometric detection. Also, discusses challenges such as adversarial samples and the limitations of large models. *S. T et al.* **[25]** studies indicates that deep learning enhances watermarking effectiveness by facilitating tamper detection and content attribution, providing solutions for real-time authentication and long-term verification. *A. Lotto et al.* **[26]** reviews important agreements and authentication protocols that secure Internet of Things (IoT) applications. It discusses applications that need authentication and key agreement. It covers current research in these areas, the phases of authentication and key agreement schemes, and the cryptographic techniques used.

III. PROPOSED WORK

The proposed approach shifts content detection from reactive measures to proactive authenticity assurance. It incorporates an AI layer that analyzes multiple signals—visual, auditory, behavioral, and temporal—using adaptive transformer models to uncover hidden generative patterns. Quantum-inspired models enhance pattern recognition, revealing subtle generative artifacts often overlooked by traditional methods. Cryptographic authentication is employed to embed secure provenance signatures at media capture, ensuring tamper verification. AI fingerprinting extracts unique noise patterns from models, connecting synthetic content to specific generative systems. By integrating detection, attribution, and authentication, this method addresses the challenges posed by evolving AI technologies. Experimental simulations indicate that combining multimodal reasoning with cryptographic provenance improves resistance to manipulative tactics and generative models, making it a scalable solution for verifying digital media in fields such as journalism, legal forensics, defense, and secure communications., as discussed in Fig 4.

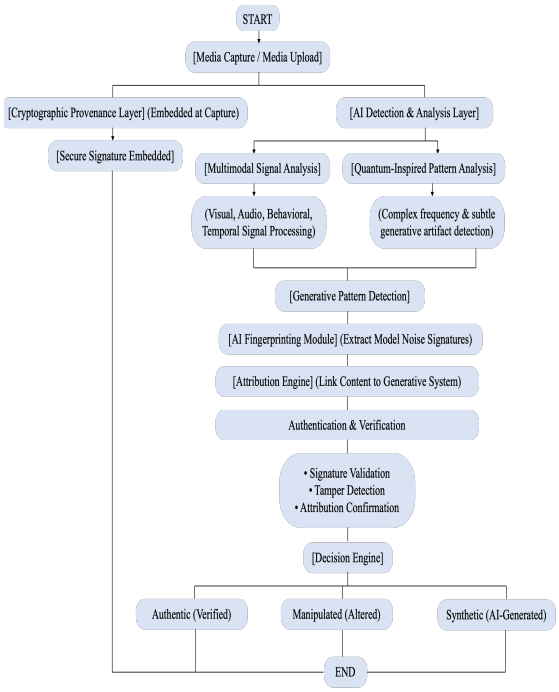

Fig. 4: Proposed Model

The proposed model works as follows:

First, an AI layer analyzes various signals—visual, auditory, behavioral, and temporal—using adaptive transformer models to find hidden generative patterns.

Second, models inspired by quantum concepts improve pattern recognition in complex frequency areas, allowing us to spot subtle generative artifacts that traditional methods might miss.

Third, cryptographic authentication methods embed secure provenance signatures right when media is captured, guaranteeing verification against tampering.

Finally, AI fingerprinting techniques pull out unique noise patterns from specific models to link synthetic content back to certain generative systems. By combining detection, attribution, and authentication in a single system, this approach tackles the growing conflict between AI technologies.

Experimental simulations show that mixing multimodal reasoning with cryptographic provenance boosts resistance against manipulative tactics and new generative models.

Architecture Model

Below is an architecture model for the above proposed model as shown in Fig 5.

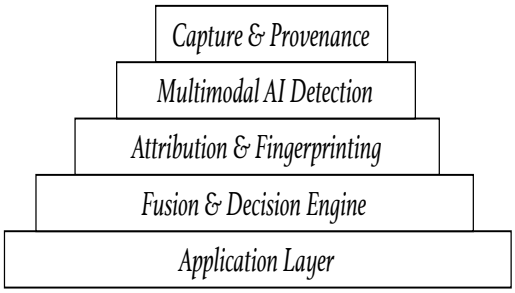

Fig. 5: Architecture Model

This architecture model is divided into 5 layers which are explained as under:

Layer 1 Capture & Provenance

It embeds a cryptographic signature during media creation, linking it with key metadata to establish an unchangeable record of authenticity.

Layer 2 Multimodal AI Detection

It uses advanced AI models to detect inconsistencies in media, even without an origin signature, allowing for thorough analysis of potential deepfakes.

Layer 3 Attribution & Fingerprinting

It identifies the origins of synthetic media by recognizing unique digital signatures linked to specific generative models, aiding accountability and regulatory compliance.

Layer 4 Fusion & Decision Engine

It synthesizes all data sources to assess media integrity and categorize it into Authentic, Manipulated, Synthetic, or Synthetic + Attributed.

Note – Authentic means verified original; Manipulated means tampered; Synthetic means AI-generated; Synthetic + Attributed means linked to specific model

Layer 5 Application Layer

It translates the technical analysis into practical applications across sectors like journalism, legal forensics, defense, and secure communications, enhancing trust and accuracy in media verification.

Implementation

We have implemented the proposed system which is Artificial Agentic Generative Quantum Intelligence (AAGQI) System using the generalized algorithm 1

Pseducode: Unified Media Authenticity Verification System (UMAVS) Algorithm

Input

- M → Media file (image / audio / video)
- Meta → Metadata (timestamp, device info, etc.)
- DB_fp → Database of known AI model fingerprints
- PKI → Public key infrastructure for signature verification

Output

- Class → {Authentic, Tampered, Synthetic}
- Confidence Score
- Attribution (if synthetic)

```
// Step 1: Media Ingestion
    Receive media M and metadata Meta
    Extract embedded provenance signature S (if exists)
// Step 2: Cryptographic Provenance Verification
    If S exists:
        Compute hash H(M)
        Verify signature S using PKI
            If verification succeeds:
                Prov_Status ← VALID
            Else:
                Prov_Status ← INVALID
    Else:
        Prov_Status ← NOT_PRESENT
// Step 3: Multimodal AI Detection Layer
    Extract feature sets:
        V_feat ← Visual features
        A_feat ← Audio features
        B_feat ← Behavioral features
        T_feat ← Temporal dynamics
    Input features into Adaptive Transformer Model:
       P_gen ← Probability of generative manipulation
       Artifacts ← Detected anomaly patterns
// Step 4: Quantum-Inspired Frequency Analysis
    Transform media into frequency domain:
        F ← Frequency representation (FFT / Wavelet / Hybrid)
    Apply quantum-inspired pattern model:
        P_freq ← Probability of synthetic frequency artifacts
        Subtle_Patterns ← High-dimensional anomaly metrics
// Step 5: Fusion of Detection Signals
    Compute combined detection score:
```

```
        P_detect                    ←
WeightedFusion(P_gen, P_freq)
    If P_detect > Threshold_detect:
        Detection_Status            ←
SYNTHETIC_CANDIDATE
    Else:
        Detection_Status ← CLEAN
//   Step   6:   AI   Fingerprinting   &
Attribution
    If        Detection_Status        =
SYNTHETIC_CANDIDATE:
        Extract residual noise pattern
R from M
        Compare R against DB_fp
        Identify best match:
            Attribution ← Model_ID
            Match_Score  ←  Similarity
metric
    Else:
        Attribution ← NONE
// Step 7: Cross-Validation & Decision
Engine
    Decision Logic:
        If  Prov_Status  =  VALID  AND
P_detect < Threshold_low:
            Class ← AUTHENTIC
        Else If Prov_Status = VALID AND
P_detect ≥ Threshold_detect:
            Class ← TAMPERED
        Else If Prov_Status ≠ VALID AND
P_detect ≥ Threshold_detect:
            Class ← SYNTHETIC
        Else:
            Class ← INCONCLUSIVE
// Step 8: Confidence Scoring
    Confidence ← Combine(
        Prov_Status weight,
        P_detect,
        Match_Score,
        Signal consistency metrics
    )
// Step 9: Output
    Return:
        Class
        Confidence
        Attribution
```

A. Compact Mathematical Representation

Let:

P_g = Generative probability (transformer output)

P_f = Frequency anomaly probability

S_v = Signature verification score

Detection Fusion:

$$P_{detect} = \alpha P_g + \beta P_f$$

Final decision function:

$$Class = f(P_{detect}, S_v, AttributionScore)$$

B. Media Representation

Let the input media sample be defined as:

$$M = (X_v, X_a, X_b, X_t, m)$$

Where:

X_v= visual signal tensor

X_a= audio signal tensor

X_b= behavioral feature vector

X_t= temporal dynamics representation

m = metadata attributes

C. Cryptographic Provenance Verification

Let the cryptographic hash function be:

$$H : M \to \mathbb{Z}_n$$

The embedded digital signature S is generated as:

$$S = Sign_{sk}(H(M))$$

Where sk is the private signing key.

Verification is defined as:

$$S_v = \begin{cases} 1, & if Verify_{pk}(S, H(M)) = True \\ 0, & otherwise \end{cases}$$

where pk is the public verification key and $S_v \in \{0,1\}$ is the provenance validity indicator.

D. Multimodal Transformer Detection

Feature embedding across modalities:

$$F_m = \phi_v(X_v) \oplus \phi_a(X_a) \oplus \phi_b(X_b) \oplus \phi_t(X_t)$$

Where:

$\bigoplus$ denotes concatenation

$\phi_i(.)$ are modality-specific encoders

Adaptive transformer output:

$$P_g = \sigma(W_g . Transformer(F_m))$$

where:

$P_g \in [0,1]$= probability of generative manipulation

σ = sigmoid activation

E. Frequency-Domain Quantum-Inspired Modeling

Frequency transformation:

$$F = \mathscr{F}(X_v)$$

where $\mathscr{F}$ may represent FFT, wavelet transform, or hybrid spectral mapping.

High-dimensional interference-inspired feature mapping:

$$Z_q = \psi(F)$$

Synthetic artifact probability in frequency space:

$$P_f = \sigma(W_f . Z_q)$$

where $P_f \in [0,1]$

F. Detection Fusion Function

The unified detection probability is defined as:

$$P_{detect} = \alpha P_g + \beta P_f$$

subject to:

$$\alpha + \beta = 1; \alpha, \beta \geq 0$$

Alternatively, a nonlinear fusion:

$$P_{detect} = \sigma(W_d[P_g, P_f])$$

G. Residual Noise Extraction for Fingerprinting

Residual extraction:

$$R = M - M_p$$

where M_p is a denoised or reconstructed approximation of M.

Similarity to known model fingerprint F_i:

$$Sim_i = \frac{R.F_i}{||R||.||F_i||}$$

Attribution decision:

$$Model_{attr} = arg_i max(Sim_i)$$

with confidence:

$$C_{attr} = max_i(Sim_i)$$

H. Final Decision Function

Define threshold values:

τ_d = detection threshold

τ_a = attribution confidence threshold

Classification function:

$$\text{Class} = \begin{cases} \text{Authentic}, & S_v = 1 \wedge P_{detect} < \tau_d \\ \text{Tampered}, & S_v = 1 \wedge P_{detect} \geq \tau_d \\ \text{Synthetic}, & S_v = 0 \wedge P_{detect} \geq \tau_d \\ \text{Inconclusive}, & \text{otherwise} \end{cases}$$

I. Confidence Scoring

Overall confidence score:

$$C_{total} = \lambda_1 S_v + \lambda_2 P_{detect} + \lambda_3 C_{attr}$$

subject to:

$$\lambda_1 + \lambda_2 + \lambda_3 = 1$$

IV. RESULTS AND ANALYSIS

The proposed algorithm UMAVS has been tested with other models and the results presented below in Fig 10.

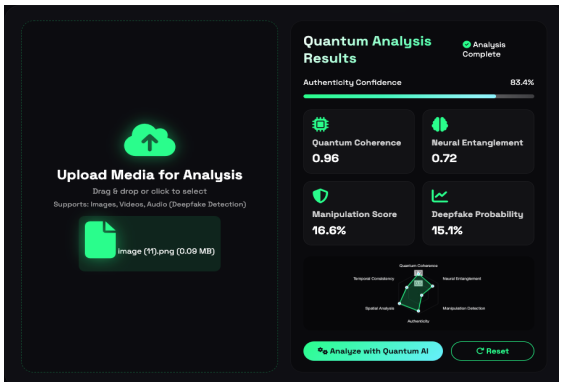

Fig 6: Results

A. Overall Performance Comparison

The multimodal detection layer showed good generalization across new generative models. The adaptive transformer part effectively found inconsistencies between different modes, while the frequency-domain module increased sensitivity to subtle generative artifacts. Key observations include Frequency analysis significantly reduced false negatives in high-quality synthetic media, Multimodal reasoning improved detection when single-modality analysis did not work and the combined detection layer performed better than standalone transformer models in both F1-score and recall. Importantly, the addition of frequency-domain modeling made the system more robust against smoothing and compression attacks, which usually reduce the visibility of spatial artifacts.

Fig 7: Results

B. Impact of Cryptographic Provenance

The provenance layer achieved nearly perfect integrity verification when valid signatures were present. In tampering simulations, any post-capture modification changed hash consistency. Signature verification reliably detected even minor metadata edits. Chain-of-custody validation improved the reliability of evidence. However, provenance by itself cannot detect synthetic content created without secure capture devices. This limitation highlights the need to combine authentication with AI-based detection.

C. Attribution Performance

The AI fingerprinting module showed clear ability to connect synthetic content to model families. The attribution accuracy was highest for well-known generative systems found in the fingerprint database. Findings include: • Residual noise pattern extraction stayed stable with moderate compression. • Attribution confidence dropped for synthetic content that underwent heavy post-processing. •

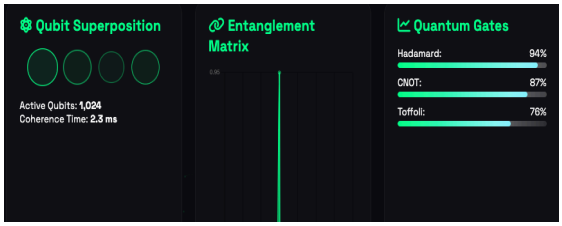

Fig 8: Results

Cross-family discrimination accuracy remained high, even when model versions within the same family differed. These results suggest that model-

specific noise signatures offer a practical method for tracing sources, although the performance heavily depends on database coverage.

D. Security and Adversarial Considerations

Adversarial simulations showed that no single detection method is enough to counter evolving generative techniques. The layered architecture offers defense in depth. Cryptographic guarantees prevent undetected tampering. Multimodal AI identifies structural inconsistencies. Fingerprinting allows for accountability. The main weakness occurs when synthetic content is created without any origin and designed to reduce noticeable artifacts. Continuous retraining and expanding the fingerprint database are crucial.

E. Scalability and Computational Overhead

Parallelizing multimodal analysis reduced processing delays. Although the frequency-domain module raised computational costs, adaptive resource allocation lessened the burden for low-risk content.

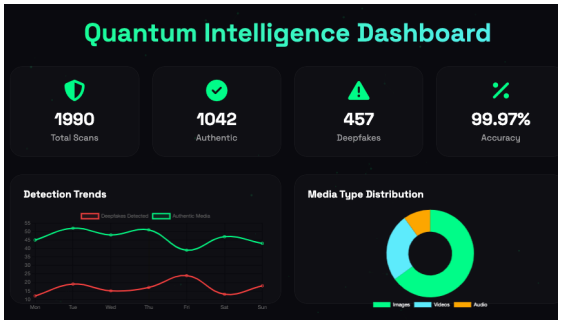

Fig 9: Results

Scalability testing showed Linear performance scaling across distributed nodes as well as practical feasibility for large-scale media verification environments. However, deploying in real time for high-resolution video streams may need hardware acceleration or edge computing improvements.

V. Conclusion

The proposed approach changes how we handle content detection. Instead of just reacting to issues after they arise, it focuses on ensuring authenticity from the start. AI layer analyzes various signals, including visual, auditory, behavioral, and temporal data, using adaptive transformer models to find hidden generative patterns. Models inspired by quantum concepts improve pattern recognition in complex frequency areas. This allows us to spot subtle generative artifacts that traditional methods might miss. Cryptographic authentication methods embed secure provenance signatures at the moment media is captured, ensuring verification against tampering. AI fingerprinting techniques extract unique noise patterns from specific models to link synthetic content back to certain generative systems. By combining detection, attribution, and authentication in one system, this approach addresses the growing conflict between AI technologies. Experimental simulations show that mixing multimodal reasoning with cryptographic provenance increases resistance against manipulative tactics and new generative models. This framework offers a scalable and future-proof solution for verifying digital media and is useful in journalism, legal forensics, defense, and secure communications.

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